

Team Members with different roles. There are no requirements in which team a Team Member is part of (so this could be outside of the central Data Team as well) as long as the TTT meets the expectations described below.

Tool / Technology Tandem	
----- -----	
Snowflake	t.b.d.
Monte Carlo	t.b.d.
dbt	t.b.d.
Tableau	t.b.d.

What do we expect from TTT

- We expect TTT to get in touch with our business partners and all functions that contribute to the data program or work with our Data Platform, to understand their challenges.
- We expect TTT to get up to date with the latest in their area. They understand the full capabilities of the tool / technology, have regular touchpoints with the respective vendor and have a good understanding of the latest released features.
- TTT will guide and educate our business partners.
- TTT will initiate design-spikes for quarterly planning.

SECTION: enterprise-data/organization/analytics-engineering.md

The mission of the Analytics Engineering team is to maximize the impact of business decisions and strategy with data solutions that are trusted and scalable.

We do this by helping all GitLab teams move up the Data maturity by means of our GitLab values and our Data Team Principles.

Analytics Engineering Responsibilities

The Analytics Engineering Team is directly responsible for:

- Integrating and conforming data from 50+ sources into a single analytics platform
- Encoding business logic into dbt data models that support enterprise-grade reporting and analytics

Additionally, the Analytics Engineering Team supports the following responsibilities

- With Data Leadership:
 - Define and publish a Data Strategy to help maximize the value of GitLab's Data Assets
 - Broadcast regular updates about data deliverables, ongoing initiatives, and upcoming plans
- With the Data Platform Team:
 - Build and maintain the company's central Enterprise Data Warehouse to support Reporting, Analysis, Dimensional Modeling, and Data Development for all functional teams
- With the Data Governance and Quality Team:
 - Help define and champion Data Quality practices and programs for GitLab data systems
- With the Data Science and Enterprise Analytics Team:

- Provide Self-Service Data capabilities to help everyone leverage data and analytics

Analytics Engineering Organization

The Analytics Engineering Organization is composed of sub-teams as follows:

- GTM Analytics Engineering
- R&D Analytics Engineering
- Finance Analytics Engineering
- People Analytics Engineering

Analytics Engineering Ceremonies

Each sub-team has the following ceremonies. They can be at a cadence that makes sense for the sub-team and can also use asynchronous or synchronous approaches as agreed upon by the sub-team. It is optional for team members to attend the ceremonies of other pillars. Team members are encouraged to attend other pillar ceremonies where attendance adds value.

- Backlog Refinement Sync
- Iteration Planning Sync
- Sub-Team Meeting
- Analytics Engineering Functional Team Sync (All Sub-Teams Attend)

Analytics Engineering Processes

- Triage follows the Enterprise Data Team Triage Process. This is the primary intake process for P1-Ops issues.

Analytics Engineering Leadership Roles

- The Manager, Data provides leadership to the sub-teams as described in the Senior Manager, Data Job Family
- Lead Analytics Engineer (Stable Counterpart for the Business)
 - The Lead Analytics Engineer fulfills the below responsibilities from the Senior Analytics Engineer Job Responsibilities:
 - Own one or more stakeholder relationship in Go To Market, Research & Development, General & Administrative, Financial Analytics, or Engineering Analytics business functions.
 - Co-DRI of Key Results along with the Manager, Data.
 - Lead work breakdown sessions for OKRs.
 - Work with functional stakeholders to prioritize P3-Other issues.
 - Serve as a primary or back-up Maintainer on the Data Team Project. Provide final review, feedback, and approval of Merge Requests.
 - Review the weekly stand-up and provide support as needed to unblock team members and answer questions.

SECTION: enterprise-data/organization/analytics.md

The mission of the Data Analytics team is to maximize the impact of business decisions and strategy with data solutions that are trusted and scalable.

We do this by helping all GitLab teams move up the Data maturity by means of our GitLab values and our Data Team Principles.

Data Analytics Responsibilities

The Data Analytics Team is directly responsible for:

- Managing and governing the company's Key Performance Indicator definitions, database, and data visualizations
- Providing customizable Data Services, including Data Visualization, Data Modeling, Data Quality, and Data Integration
- Supporting the company's governance, risk, and compliance programs as they relate to Data & Analytics systems

Additionally, the Data Analytics Team supports the following responsibilities

- With the Data Leadership Team:
 - Defining and publishing a Data Strategy to help maximize the value of GitLab's Data Assets
 - Broadcasting regular updates about data deliverables, ongoing initiatives, and upcoming plans
- With the Data Platform Team:
 - Building and maintaining the company's central Enterprise Data Warehouse to support Reporting, Analysis, Dimensional Modeling, and Data Development for all GitLab teams
 - Developing Data Management features such as master data, reference data, data quality, data catalog, and data publishing
 - Providing Self-Service Data capabilities to help everyone leverage data and analytics
 - Helping to define and champion Data Quality practices and programs for GitLab data systems

Handbook First on the Data Analytics Team

At GitLab we are Handbook First and drive this by:

1. Helping our Business Partners (all GitLab team members) update the handbook with the most up-to-date and accurate information.
1. Reviewing content and ensure that the information in the handbook best represents the data, taking into account data sensitivity and security.
1. Updating the Data Team handbook section
 - with exciting Data Analytics content
 - Examples of upcoming content: Data Visualization: Tips and Tricks and Data Storytelling
 - to reflect changes in Data Team process or codebase (and yes we get very excited about better processes and code! Who doesn't!)

What Is Analytics And Why Do We Do It?

Analytics is the practice of turning data into information, knowledge, and wisdom to:

1. Improve Efficiency & Avoid Costs
 1. Human Capital - Location Factor, Offer Accept Rate, Cost Per Hire
 1. Finance - ARR, New Pipeline Created, CC Failure Rate
 1. Sales Channel - Win Rate, Services Attach Rate,
 1. Marketing - Opportunities Created, Product Downloads
1. Improve the Business Model
 1. Analytics Instrumentation - Sign-up Growth, New Features Delivered, Top Features
 1. Customer Analytics - Firmographics, Adoption Trends, Usage vs. Subscription
 1. Customer Purchasing Trends - Renewals, Upgrades, Downgrades
1. Improve the Customer Experience
 1. Data Products - Industry Benchmarks, Data APIs, Algorithms
 1. Data-Informed Process Improvements
 1. In-Product Insights

Data Analysis Process

Data Analytics, Advanced Data Analytics, and Data Science all start with the Data Analysis Process.

The Data Analysis Process has five steps:

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graph LR

A[Problem Statement]-->B

B[Data Collection]-->C

C[Pre-Processing]-->D

D[Modeling]-->E

E[Presentation]-->A

In Data Analytics, context is everything. It guides the way Data Analyst view the problem, the data, and the methodology taken to generate the data insights.

Problem Statement

In the Problem Statement step, it is crucial that we define the business problem with the Business Partner. By clearly defining the business problem that the Business Partner is seeking data insights into, Data Analyst can set success criterias to ensure that the problem is technically solvable with the data available and to provide the maximum business insight into the problem.

Data Collection

In the Data Collection step, it's important for Data Analyst to explore and understand the Data Caveats that may limit the impact or generalization of the analysis as well as to evaluate any Data Biases or Data Concerns. Understanding how the data is collected in each business system will ensure that we are reporting on accurate data in an appropriate manner. During this step, it's important to ensure that the Data Types are accurate for reliable transformations.

Therefore, this step is actually a part of a feedback loop with the source system owners. If a Data Team member finds that a data field is missing or inaccurately captured, it is the responsibility of anyone at GitLab to reach out to the source system owners and ask them to update or add the data field in question.

Pre-Processing

In the Pre-Processing step, Data Analyst prepare and cleanse the data and exam the data quality to design the optimal Data Structures for scalable reporting. It's important in this step to review the model and ensure that the model is business-friendly.

Modeling

In the Modeling step, we aim to create new data models in the Enterprise Dimensional Model format for all GitLab team members. We start first with an Entity Relationship Diagram (ERD) to ensure that the new data structures reflect accurate business processes. All data models are reviewed by the Data Platform Team.

Presentation

In the Presentation step, Data Analyst radiate in their Data Storytelling skills with strong Data Visualizations. Data Analyst recognizes that it is important to package and display the actionable business insights to effectively communicate with stakeholders. In fact, it is as vital as ensuring that the data solution, such as a Business Intelligence (BI) dashboards with drill-downs to show the granular reasoning behind the insights, has robust data quality checks for our Business Partners (all GitLab team members) to trust the data.

The wonderful thing about Data Analytics is that each insight should propagate a wave of additional business questions to solve, which then allows the Data Analysis Process to cycle again.

SECTION: enterprise-data/organization/data-steering.md

Data Steering Committee Purpose

The purpose of this Data Steering Committee is to oversee and drive the strategic direction of GitLab Enterprise data initiatives. The forum ensures that data is leveraged effectively to support business goals, improve decision-making processes, and drive innovation. It acts as a governing body to help establish policies, standards, and best practices for data governance, data quality, data privacy, data adoption, and data security.

Data Steering Committee Charter

The Data Steering Committee Charter sets out the broad expectations on the responsibilities and outcomes achieved by the committee. It tries to set the meeting format, participants and desired agenda across broad data management topics.

Scope

The enterprise data initiatives/projects should broadly be under the scope of following categories:

1. Data Strategy
1. Data Quality Management
1. Data Security
1. Data Lifecycle Management
1. Data Architecture and Modeling
1. Data Discovery/Metadata Management
1. Data Mastery and Reference Data
1. Data Adoption/Data Literacy
1. Data as value driver and outcome oriented

Responsibilities

- Define, approve and communicate enterprise data strategies, data policies, data standards, procedures and metrics.
- Tracks and enforces regulatory compliance and conformance to data policies, standards and procedures
- Understand and promote the value of data assets
- Initiate, track and oversee the delivery of committed quarterly KRs
- Important decision making to bring consensus around data processes, enterprise metric definitions, data mastery or reference data and validation procedures

Composition

The Data Steering Committee consists of

- Head of Data & Insights
- Leadership team of Data & Insights
- Senior leadership from lines of business
- Senior leadership from IT
- Senior leadership from Security, Legal and Audit
- Data Governance Program Manager

Subcommittees or Working Groups · The Data Steering Committee may form and delegate authority to subcommittees, consisting of one or more members of the overall Committee. These subcommittees would be responsible for evaluating and providing recommendations for specific topics before the Committee.

Meetings and Communication

Most of the communication will be done ASYNC through slack channel.

The meetings will be scheduled on a need basis. The meeting will be chaired by the Head of Data & Insights. Meetings may be called by the Chair as needed. Any member may propose additional meetings by contacting the Chair, if needed.

Measuring and Reporting

The following reporting will be built (in future, targeting H2 FY26) to track the progress and

success of the initiatives -

- Program Status Reporting
- Data Quality Dashboard
- Data Asset Library

Charter Review

This charter will be reviewed every year by Data Steering Committee to ensure it continues to meet the needs and includes the evolving best practices.

SECTION: enterprise-data/organization/engineering.md

The mission of the Data Platform team is to build a secure and trusted data platform that makes it possible for everyone to be an analyst so that our only limitations are the data or analysts themselves. We do this by means of our GitLab values and our Data Team Principles

Data Platform Responsibilities

The Data Platform team is directly responsible for:

- Integrating new data sources to enable analysis of subject areas, activities, and processes
- Keeping the data sources running per established Service Level Objectives and Tiers
- Expanding the scalability of pipelines and processes to meet growing data volumes
- Evaluating new technologies to help make our platform more scalable and efficient

Handbook First on the Data Platform Team

At GitLab we are Handbook First. Any changes to our codebase or process should have a handbook MR created before the work is started or concurrently with the MR to change the code.

For example: If we are adding a data source, we would first create the merge request to update our the extract and load and system diagram sections of our handbook. This MR can be updated as the change is created and merged along with the change.

Data Platform Team knowledge sessions

We as a Data Platform Team are always eager to gain more knowledge and looking for the latest solutions available. Therefore we allocate (half of) our Data Platform Team meeting every 2 months to invite external speakers. The Data Platform Team meeting is every Tuesday at 2PM UTC and we invite any solution provider to join every first Tuesday of every even month to talk with us about Data and Technology. Feel free to reach out to us pro-actively to join 1 of our Team meetings by opening an issue or contact any of the Data Platform Team members (by email or LinkedIn).

Date	Discussion topic
2022-10-04	t.b.d.

2022-12-06	t.b.d.
2023-02-01	t.b.d.
2023-04-05	t.b.d.
2023-06-07	t.b.d.

Data Platform Team Roles

In order to efficiently operate as a team we designate and assign the following responsibilities to members of the Data Platform team. These assignments are not permanent, but do create a Directly Responsible Individual on the team for these responsibilities. The assigned Engineer is responsible for both maintenance and iterative improvement over these responsibilities.

Production Operations

Production operations is always our top standing priority on the Data Platform team and includes all technical operations that support or affect our production environment and the SLOs it supports. This includes, but is not limited to monitoring, testing, deployment, code reviews, and DevOps generally.

Project Management and Planning

The Engineer assigned the Project Management and Planning responsibility oversees, directs, and improves our workflow processes, iteration planning, and issue triaging. They ensure that work is getting planned and completed efficiently and that the work being done appropriately supports and enables the goals of both the Data Team and GitLab.

User Experience

This engineer is responsible for the usability of our data platform. As stated in the Data Platform mission, we are building the platform for everyone. The assigned Engineer proactively finds and solutions around our data platform with special mind to those experiences that are the most difficult to use.

Security, Compliance & Privacy

This role keeps our platform secure, private, and compliant and will be the point of contact for audit questions as well as the primary engineer responsible for reviewing changes that affect the security and privacy of our data platform. They should also be iteratively improving the security features of our platform.

Data Warehouse Architecture

In order to keep our iterations inexpensive and increase the usability of our warehouse we have decided to implement a dimensional warehouse. This engineer is responsible for protecting the integrity of our warehouse architecture and leads the vision on where the architecture needs to be improved. All of these changes and protections should keep our customers, both internal and external, as their primary focus.

SECTION: enterprise-data/organization/internships.md

*<i id="biz-tech-icons" class="far fa-paper-plane"></i>*Introduction

Welcome to the Data Team Internship Program page. The Program is launching in 2022 and we are excited to help interested GitLab team members experience life in the fast-paced, results-oriented, and rapidly-evolving world of Data!

- Slack Channel: data
- Internship for Learning
- Data Onboarding Issue

<i class="far fa-newspaper" id="biz-tech-icons"></i> The 2022 Data Team Internship Program

The initial focus of the 2022 Program will be Data Internship for Learning, following the GitLab Internship for Learning Program and focused on supporting career development for existing GitLab Team Members. Later in 2022 we will evaluate extending the program to support non-GitLab Team Members and traditional undergraduate internships.

How Does Data Internship For Learning Work?

The Data Internship for Learning follows GitLab Internship for Learning Program guidelines. In addition, given the rapid growth of the Data Team, we seek to partner every Intern with a current individual contributor Data Team member who is interested in exploring their own career development towards a Leadership role, such as a Team Lead specialization or Data Management role. In this model, both team members share a commitment to support each other's success.

At launch in January 2022, Data Internship for Learning will be structured as follows:

- Follows the GitLab Internship for Learning Program
- Open to only current GitLab Team Members
- Intern time allocation should range from 25%-50%, with a 3-month commitment
- Data Team Mentor time allocation should range from 15-25%, with a 3-month commitment

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subgraph "Shared Responsibility and Commitment"

A[Intern] --> B

B[Mentor] --> A

end

How Do I Get Started?

Follow the GitLab Internship for Learning Program instructions.

Criteria For Creating A Data Team Internship

Requests to create a Data Team Intern position will be evaluated on the following criteria:

1. Team Size, Maturity, and Need

- How established is the team the intern will be joining? Is this an established team with processes, roles, a roadmap, and backlog?
- Will the team be able to take on an intern without risking a decrease in velocity?
- Would the team benefit from an intern to support a more senior member's career development, to introduce new perspectives, or to accelerate the roadmap in one or more areas?
- Does the team have a roadmap containing low weight issues with few dependencies suitable for an intern?

1. Mentor Commitment

- Do the mentors have previous experience mentoring interns or junior engineers? Previous experience is a nice-to-have, but not a must-have.
- Are the mentors interested in Leadership development?
- Can the mentor commit to a 3-month engagement, including regular meetings and progress reviews? Are there potentially extended out-of-office periods that would affect the internship experience?

1. Intern Commitment

- Can the intern commit to a 3-month engagement, including regular meetings and progress reviews? Are there potentially extended out-of-office periods that would affect the internship experience?

If the answer to these questions is YES, there is a great chance the internship is a fit for all team members.

Data Team Internship KPIs

The following KPIs are guidelines we plan to use for the internship. These may be adjusted by the mentor/intern prior to the start of the internship with approval from the team manager.

1. Successful Data Team Onboarding within 30 Calendar Days
1. Monthly Progress Report and Feedback issue
1. Final CSAT Score of 4 or higher on a 5 point scale
1. Number of Data Project MRs successfully Merged > 5
1. Number of Data Team Handbook MRs successfully Merged > 5

What Does A Data Team Intern Do?

An internship gives you insight into a Data Team role by providing you opportunities to learn and contribute to GitLab results. You will be fully integrated into the team through Data Onboarding and be assigned real projects that contribute to Quarterly Objectives and the Data Team Roadmap. In the GitLab Data Team we believe in full end-to-end ownership of the work we do, ranging from requirements gathering all the way to production deployment and operations. During your internship you will be assigned a Senior team member as a mentor-manager, who will help you to navigate in daily tasks. Examples of your tasks will include:

- Requirements gathering
- Extraction of data from various sources to support project's objective

- Analysing and visualising data to develop new insights
- Collaboration with stakeholders (including explaining technical concepts with non-technical language)
- Working in GitLab to manage projects (via epics and issues)
- Writing new source code and merging the code through a Merge Request

Data Team Internship for Learning Positions

Analytics Engineering Internship for Learning

Openings

Quarter	Openings	Filled	Remaining Openings
----	----	----	----
FY23Q1	1	1	0
FY23Q2	2	2	0
FY23Q3	TBD	TBD	TBD
FY23Q4	TBD	TBD	TBD

Requirements

- Familiarity with and Interest in the Analytics Engineering job family
- BS or MS in data-related fields (statistics, computer science, engineering), but self-learners are also welcome to apply
- Positive and solution-oriented mindset
- Comfort working in a highly agile, intensely iterative environment
- Self-motivated and self-managing, with task organizational skills
- Working knowledge of SQL
- Basic knowledge of dashboard tools (we use Tableau)
- At least 1 completed demoable data project (Non-commercial projects done in a spare time also count, as do school projects)
- Nice to have
 - Understanding of software industry
 - Prior experience in a Data Role
 - Experience designing and managing projects
 - Working knowledge of Python (we use Jupyter notebooks to analyse and visualise data)

Data Science Internship for Learning

Openings

Quarter	Openings	Filled	Remaining Openings
----	----	----	----
FY23Q1	2	2	0
FY23Q2	2	2	0
FY23Q3	2	2	0
FY23Q4	3	3	0
FY24Q1	1	1	0
FY24Q2	1	1	0
FY24Q3	2	0	2

Requirements

- Familiarity with and Interest in the Data Science job family
- BS or MS in data-related fields (statistics, computer science, engineering), but self-learners are also welcome to apply
- Working knowledge of SQL
- Working knowledge of Python (we use Jupyter notebooks to analyse and visualise data)
- Basic knowledge of dashboard tools (we use Sisense)
- Understanding of basic statistics concepts. if you need a refresher, this Coursera Stanford course is a great resource
- Understanding of machine learning algorithms including decision trees, gradient and regression models
- Ability to communicate in English, including concise writing that summarises your work
- Ability to work asynchronously - as a fully-remote company we have team members all around a globe with differing working hours
- At least 1 completed data science project (Non-commercial projects done in a spare time also count, as do school projects)
- Being proactive in a work you do, but also not afraid to ask questions
- Nice to have
 - Understanding of CRISP-DM framework
 - Understanding of software industry

Data Platform Team Internship for Learning

Openings

Quarter	Openings	Filled	Remaining Openings
FY23Q1	0	0	0
FY23Q2	0	0	0
FY23Q3	0	0	0
FY23Q4	1	1	0
FY24Q1	0	0	0
FY24Q2	0	0	0
FY24Q3	0	0	0
FY24Q4	TBD	TBD	TBD

Requirements

- Familiarity with and Interest in the Data Engineering job family
- BS or MS in data-related fields (statistics, computer science, engineering), but self-learners are also welcome to apply
- Positive and solution-oriented mindset
- Comfort working in a highly agile, intensely iterative environment
- Self-motivated and self-managing, with task organizational skills
- Working knowledge of SQL
- Nice to have
 - Understanding of software industry
 - Prior experience in a Data Role
 - Experience designing and managing projects

Data Analyst Internship for Learning

Openings

No openings at this time.

SECTION: enterprise-data/organization/learning-library.md

Powered by GitLab Team Members

- How Data Teams Do More With Less By Adopting Software Engineering Best Practices - Thomas's talk at the 2018 DataEngConf in NYC
- Slides from a similar talk by Taylor at the 2019 Music City Tech Conference in Nashville
- Locally Optimistic AMA: August 2019 with Taylor Murphy
- 4 Examples of the power of open source analytics
- Deploying your first dbt project with GitLab CI
- DataOps in a Cloud Native World
- The Three Levels of Data Analysis- A Framework for Assessing Data Organization Maturity
- How to Implement DataOps using GitLab
- Lessons learned managing the GitLab Data team
- Views on Vue Podcast with Jacob Schatz and Taylor Murphy
- How to do DataOps with GitLab - Customer Call (GitLab Internal)
- GitLab for ML - Customer Call (GitLab Internal)

Recommended Reading, Listening, Watching

- The AI Hierarchy of Needs
- Data Meta Metrics
- Engineers Shouldn't Write ETL
- The Startup Founder's Guide to Analytics
- Functional Data Engineering...a modern paradigm for batch data processing
- Keep it SQL Stupid, a talk by Connor McArthur of Fishtown Analytics at DataEngConf SF '18 explaining dbt
- DevOps for AI
- What Can Data Scientists Learn from DevOps
- One Analyst's Guide for Going from Good to Great
- The Value of Data: Part 1, Part 2, Part 3
- Building a Data Practice
- Does my Startup Data Team Need a Data Engineer
- Data Science is different now (Note: this is why GitLab doesn't have a Data Scientist yet.)
- Why You Don't Need Data Scientists
- Resources Written by dbt Community Members
- Is your company too dumb to be data-driven?
- What does "self-serve" analytics mean to you?
- Models for integrating data science teams within organizations
- Building a Mature Analytics Workflow (Note: this explains the "Analytics is a subfield of

software engineering" premise.)

- DataOps playlist on YouTube

Data Newsletters

- Algos & Ethics
- Calogica
- The Carpentries
- DataEng Weekly
- Data Elixir
- Data is Plural
- Data Science Roundup Newsletter
- Data Science Weekly
- Music and Tech
- Normcore Tech
- NumLock News
- One Shot Learning
- SF Data

Data Blogs

- Airbnb
- Ask Good Questions
- Buffer Blog
- Calogica
- Fishtown Analytics Blog
- Go Data Driven
- MBA Mondays
- Mode Analytics Blog
- Multithreaded
- Tableau Data Blog
- Silota
- Wes McKinney Blog
- Data Ops
- Retina AI Blog
- StitchFix Algorithms Blog
- Five Thirty Eight
- Data.gov

Data Visualization Resources

- Storytelling with Data
- Data Revelations
- Data Visualization Catalogue
- Eager Eyes
- FiveThirtyEight's DataLab
- Flowing Data
- From Data to Viz
- Gravy Anecdote
- JunkCharts

- Make a Powerful Point
- Makeover Monday
- Perceptual Edge
- PolicyViz
- The Pudding
- Visualising Data
- VizWiz
- WTF Visualizations

Data Slack Communities

- Data Viz Society
- Data Science Community
- dbt
- Great Expectations
- Locally Optimistic
- Measure
- Meltano
- Open Data Community
- Pachyderm
- Prefect
- PyCarolinass
- R for Data Analysis
- Software Engineering Daily
- The Data School

Technical Learning Resources

- Chris Albon
- Mode SQL Tutorial
- dbt Tutorial
- Technically.dev Post on SQL
- Elements of Data Science
- Machine Learning Resources (GitLab Internal)
- Codecademy
- DataQuest
- Khan Academy
- HackerRank (Exercises)
- Udacity
- Stanford University Mini-Courses
- The Data School by Chartio
- W3Schools

SECTION: enterprise-data/organization/principles.md

SECTION: enterprise-data/organization/data-science/_index.md

{{% alert title="Purpose" color="success" %}}

This page is focused on the operations of GitLab's internal Enterprise Data Science Team. For information about GitLab's Product Data Science Capabilities, please visit GitLab ModelOps

{{% /alert %}}

The Enterprise Data Science Team at GitLab

The mission of the Data Science Team is to facilitate making better decisions faster using predictive analytics.

Handbook First

At GitLab we are Handbook First and promote this concept by ensuring the data science team page remains updated with the most accurate information regarding data science objectives, processes, and projects. We also strive to keep the handbook updated with useful resources and our data science toolset.

Learning About Data Science

Check out this brief overview of what data science is at GitLab:

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<figure class="videocontainer">

<iframe src="https://www.youtube.com/embed/wRbNExL0hv8" frameborder="0" allowfullscreen="true"> </iframe>

</figure>

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(Corresponding slides)

AMAs:

- 2021-09-15 AMA Recording, Presentation,
- 2021-12-09 AMA Recording, Presentation

{{% alert title="Want to Learn More?" color="success" %}}

Visit Slack data-science, watch a Data Team video. We want to hear from you!

{{% /alert %}}

Common Data Science Terms

- Accuracy - ability of a Data Science model to capture all correct data points out of all possible data points
- Algorithm - sequence of computer-implementable instructions used to solve specific problem
- Classification - process of predicting a category for each observation. For example, determining if a picture is of a cat or a dog
- Clustering - process of finding natural groupings of observations in dataset. Often used for

segmentation of users or customers

- Data Science (DS) - interdisciplinary field that uses computer science, statistical techniques and domain expertise to extract insights from data
- Exploratory Data Analysis (EDA) - analysis of data that summarizes its main characteristics (includes statistics and data visualisation)
- Feature - single column in dataset that can be used for analysis, such as country or age. Also referred to as variables or attributes
- Feature Engineering - process of selecting, combining and transforming data into features that can be used by machine learning algorithms
- Imputation - process of replacing missing or incorrect data with statistical "best guesses" of the actual values
- Machine Learning (ML) - use and development of algorithms, without being explicitly programmed to determine patterns in data
- Model - a complex set of mathematical formulas that generates predictions
- Propensity modeling - building models to predict specific events by analyzing past behaviors of a target audience.
- Regression - a statistical method for predicting an outcome. For example, predicting a person's income, or how likely a customer is to churn
- Scoring - process of generating predictions for the new dataset
- Training - process of applying an algorithm to data to create a model
- Test Dataset - deliberately excluding some observations from training the model so they can be used to verify how well the model predicts
- Weight - numerical value assigned to feature that determines its strength

Data Science Responsibilities

The Data Science Team is directly responsible for:

- Delivering descriptive, predictive, and prescriptive solutions that promote and improve GitLab's KPIs
- Being a Center of Excellence for predictive analytics and supporting other teams in their data science endeavors
- Developing tooling, processes, and best practices for data science and machine learning

Additionally, the Data Science Team supports the following responsibilities:

- With Data Leadership:
 - Scoping and executing a data science strategy that directly impacts business KPIs
 - Broadcasting regular updates about deliverables, ongoing initiatives, and roadmap
- With the Data Platform Team:
 - Defining and championing data quality best practices and programs for GitLab data systems
 - Deploying data science models, ensuring data quality and integrity, shaping datasets to be compatible with machine learning, and bringing new datasets online
 - Creating data science pipelines that work natively with the GitLab platform and the Data Team tech stack
- With the Data Analytics Team:
 - Incorporating data science into analytics initiatives
 - Designing dashboard to enhance the value and impact of the data science models

How We Work

As a Center of Excellence, the data science team is focused on working collaboratively with other teams in the organization. This means our stakeholders and executive sponsors are usually in other parts of the business (e.g. Sales, Marketing). Working closely with these other teams, we craft a project plan that aligns to their business needs, objectives, and priorities. This usually involves working closely with functional analysts within those teams to understand the data, the insights from prior analyses, and implementation hurdles.

The Data Science flywheel is focused on improving business efficiency and KPIs by creating accurate and reliable predictions. This is done in collaboration with Functional Analytics Center of Excellence to ensure the most relevant data sources are utilized, business objectives are met, and results can be quantifiably measured. As business needs change, and as the user-base grows, this flywheel approach will allow the data science team to quickly adapt, iterate, and improve machine learning models.

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graph BT;

id1(Faster, More Accurate Predictions)-->id2(Increased Business understanding) &
id5(Continuous Feedback)

id2-->id3(More Revenue & Users)

id5-->id1

id3-->id4(More Data)

id4-->id1

Data Science Initiatives

Examples of current Data Science initiatives include:

- Revenue Expansion
- Churn Reduction
- Improved Forecasting
- Customer Health
- MLOps with GitLab

Please refer to the Data Science Initiatives Internal Handbook for up-to-date information on all our on-going and planned projects.

Project Structure

The Data Science Team follows Cross-Industry standard process for data mining (CRISP-DM), which consists of 6 iterative phases:

1. Business Understanding

- Includes requirements gathering, stakeholders interviews, project definition, product user stories, and potential use cases in order to establish success criteria of the project.

1. Data Understanding

- Requires determining the breadth and scope of existing relevant data sources. Data scientists work closely with data engineers and data analysts to determine where gaps may exist and to identify any data discrepancies or risks.

1. Data Preparation

- Requires conducting data quality checks and exploratory data analysis (EDA) to develop a greater understanding of data and how different datapoints relate to solving the business need.

1. Modeling

- Machine learning techniques are used to find a solution that addresses the business need. This often takes the form of predicting why/when/how future instances of a business outcome will occur.

1. Evaluation

- Performance is generally measured by how accurate, powerful, and explainable the model is. Findings are presented to the stakeholders for feedback.

1. Deployment

- Once the model has been approved it then gets deployed into the data science production pipeline. This process automatically updates, generates predictions, and monitors the model on a regular cadence.

The GitLab approach

The Data Science Team approach to model development is centered around GitLab's value of iteration and the CRISP-DM standard. Our process expands on some of the 6 phases outlined in CRISP-DM in order to best address the needs of our specific business objectives and data infrastructure.

Data Science Platform

Our current platform consists of:

- the Enterprise Data Warehouse for storing raw and normalized source data as well as final model output for consumption by downstream consumers
- JupyterLab for model training, tuning, and selection
- GitLab for collaboration, project versioning, and source code management, experiment tracking, and CI/CD
- GitLab CI for automation and orchestration
- Monte Carlo for drift detection
- Tableau Server for model monitoring and on-going performance evaluation
- Feast as an open-source Feature Store for Machine Learning models

Current State Data Flows

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graph

```
A[Enterprise Data Warehouse: Raw and Normalized Data Sources]
B[JupyterLab & GitLab CI/CD: Model Training, Tuning, and Selection]
C[GitLab CI/CD & Pipeline Schedules: Batch scoring with Papermill]
F[Enterprise Data Warehouse: Model Output for Consumption]
D[Salesforce/Marketo: CRM Use Cases]
E[Tableau/Monte Carlo: Model Monitoring and Reporting]
G[GitLab: Source Code Management]
H[Experiment tracking]
A --> |ODBC| B
B --> H
H --> B
B --> G
G --> B
G --> C
C --> |JSON| F
F --> |CSV| D
F --> |ODBC| E
```

Feast: Feature Store Implementation

We use Feast as an open-source Feature Store for our machine learning models. Configuration can be found on the Feast project repository, updating the feature store is done using GitLab CI/CD and the web UI is published in a VM on GCP.

You can use the following pages to find more details on:

1. How to use Feast to fetch features to train and deploy Machine Learning models.
1. Feast - Feature Store Implementation Internal handbook section.

CI/CD Pipelines for Data Science

We deploy all of our models using the native GitLab CI/CD capabilities. Please see Getting Started With CI/CD for Data Science Pipelines for the most up-to-date information and instructions on how to deploy models with CI/CD

Data Science Tools at GitLab

- Pre-configured Data Science Environment: The data science team uses JupyterLab pre-configured with common python modules (pandas, numpy, etc.), native Snowflake connectivity, and git support. Working from a common framework allows us to create models and derive insights faster. This setup is freely available for anyone to use. Check out our Jupyter Guide for additional information.
- GitLab Data Science Tools for Python: Functions to help automate common data prep (dummy coding, outlier detection, variable reduction, etc.) and modeling tasks (i.e. evaluating model performance). Install directly via pypi (pip install gitlabds), or use as part of the above Data Science Environment.
- Modeling Templates: The data science team has created modeling templates to allow you to

easily start building predictive models without writing python code from scratch. To enable these templates, follow the instructions on the Jupyter Guide.

SECTION: enterprise-data/organization/data-science/project_dev_approach/index.md

Our Approach

The Data Science Team approach to model development is centered around GitLab's value of iteration and the CRISP-DM standard. Our process expands on some of the 6 phases outlined in CRISP-DM in order to best address the needs of our specific business objectives and data infrastructure:

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graph

```
A[Requirements Gathering]
B[Review of Prior Analyses]
C[Review Data Sources and Exploratory Data Analysis]
D[Feature Engineering]
E[Modeling & Implementation Plan]
F[Train MVP Model]
G[Present Model Results and Insights to Stakeholders]
I[Operationalize Model]
J[Model Monitoring]
K[Project Retrospective]
A --> B
B --> C
C --> D
D --> E
E --> |Iterative| C
E --> F
F --> G
G --> I
I --> J
J --> K
K --> A
```

- Recommended check-ins with stakeholders/project owner at the following phases, at minimum, to ensure project is on target for achieving project objectives:

- Requirements Gathering
- Modeling & Implementation Plan
- Present Model Results and Insights to Stakeholders

- For definitions of some of the terms used, please refer to Common Data Science Terms. For definitions around sizing, please see T-Shirt Sizing Approach.

- Create a new issue using the Data Science Process Template

1: Requirements Gathering

Sizing: Small (with on-going refinements through Implementation Plan Phase)

Purpose: This is perhaps the most critical phase in any data science project. Often a stakeholder will have a general idea of the problem they want to solve and you will need to help them define and refine the scope of the project before beginning to develop a modeling and implementation strategy.

Tasks:

- Schedule kick-off call with stakeholders
- Questions to be answered during kick-off call
 - What is the problem statement that is trying to be solved/answered?
 - This often starts off as very generic question, e.g., "How can we reduce churn?". It is our job to narrow in and clearly understand the population, outcome, and timeframe of the problem statement. For example, are we interested in all churning customers? How is churn defined? And over what period of time?
 - Push the stakeholders to give as many specifics as possible. This may take several meetings and some data exploration to pin down. In some cases, there may not be clearly established definition and it will be up to the data scientist to create the definition during the EDA Phase.
 - What do you hope to be able to achieve at the completion of this project?
 - The intent of this question is to get at how this project can make a meaningful impact to the problem statement. Data science projects can increase the understanding of a problem statement (e.g. understanding churn risks), but they also should have a clear strategy for making actionable change to the problem (e.g. reducing churn). If a project is purely about understanding then it is probably not an appropriate scope for a data science project.
 - Some good follow-up questions to this may be: "How do you envision the outcomes of this project to be used?"; "Who is the intended consumer of outputs from this project?"
 - Has any analysis been done previously to try and address this problem/question?
 - If prior analysis exists, this is a good indication that data exists to answer the question.
 - Who is the best technical contact to work with on reviewing any prior analyses and data sources?
- Schedule a reoccurring meeting cadence with stakeholders. This should probably be more frequent at first (i.e. weekly) and can then move to a less frequent cadence (i.e. every other week) as the project progresses

Considerations:

- Early in the conversation with the stakeholders it is critical to understand if the project is the appropriate scope for data science and if there appears to be the relevant data to answer the problem statement.

Completion Criteria:

- Project determined to be in the appropriate scope for a data science project
- On its face, it appears the data exists to answer the problem statement

2: Review of Prior Analyses (Optional)

Sizing: Small

Purpose: understanding previous analytic work is essential for developing efficient and effective data science projects. By knowing the work that has already been done we can identify useful data sources, outcome (target) definitions, potential predictors (features), nuances in the data, and important insights.

Tasks:

- Schedule a call with the DRI of the prior analytics work
- understand what data sources were used (and why).
- understand what data sources were considered but not used (and why)
 - Also includes understanding if there are any potentially useful data sources that were not used (and why)
- Any technical/syntax definitions that should be used for this project.
 - Links to any relevant code
- What were the key insights and takeaways from the analysis?
 - Links to decks and/or documentation
- If the analysis was repeated today, what would be different (and why)?

Considerations:

- Prior analyses are generally helpful to review to avoid duplicating efforts and to have a base of knowledge to build from. Consider leveraging pieces of the analysis code (if available) as part of the EDA and Feature Engineering Phases.

Completion Criteria:

- A solid understanding of the analysis based on the Tasks above

3a: Review Data Sources and Exploratory Data Analysis (EDA)

Iterative with 3b & 3c

Sizing: Large

Purpose: Review available relevant data and conduct analysis around the outcome/target and potential predictors (features). This will allow you to narrow in on the necessary data sources to be used in the Train MVP Model phase. It is important to understand how your outcome/target relates to your potential predictor data and to set up the prediction timeframe appropriately.

Tasks:

- understand if we have data available to measure the intended outcome/target of the project.
 - This is the most critical element to establish before continuing. If we cannot quantify and

measure the outcome, we cannot predict future instances of it.

- E.g. Churn requires an examination of historical revenue figures to determine when a customer reduced spend. Therefore, historical snapshots of ARR data would need to exist for a sufficient period in order to build predictions. As a general rule, at least twice the amount of historical data will need to exist as your prediction timeframe. For example, if you are predicting churn in the next 12 months, then at least 24 months of revenue data will need to exist (12 months for your prediction window and 12 months prior to that window to capture predictors/features).
- Determine if there are sufficient instances of the outcome/target that would make it reasonable to build a model on.
 - A very, very general rule of thumb: At least 1,000 instances of the outcome/target
- Review all relevant data sources that you intend to use for modeling
 - Do the data sources contain data from the time periods you are interesting in examining?
 - Do the data sources contain sufficient coverage? That is, are there are lot of missing values or features with largely the same value for every row?
 - What is the relationship (correlation) between the data and the outcome/target? If the relationship is too strong or too weak, it may not be a good candidate to include in the modeling phase.
 - Are there are potential risks for data leakage) from the lookback window into the prediction window by using this data source?

Considerations:

- It is not necessary to go too deep into creating features at this phase. What is important is to get a sense if the data source will be useful and proper for modeling the outcome/target. It may also be helpful to start constructing the framework your ETL code (most likely SQL) during this phase.

Completion Criteria:

- Generate syntax for outcome/target and verify sufficient count exists for modeling
- Identify possible datasources and determine viability of each source for use in the project
- Perform exploratory data analysis
- Review results with stakeholders and collect feedback on appropriateness of:
 - outcome/target definition
 - data sources being used, reasons for inclusion/exclusion, and if any additional sources should be considered.
 - Solicit feedback regarding fields that are being created as potential features of the model. Stakeholders may suggest addition datasources, features, or rollups that are not currently being considered.
 - This will likely be an iterative process. As you uncover insights, it may unearth additional questions/comments/concerns from stakeholders.

3b: Feature Engineering

Iterative with 3a & 3c

Sizing: Large

Purpose: Based on the EDA, prior analysis, and knowledge about the problem statement, create a

list of features in SQL using the tables outlined in the implementation plan that will be used to predict the outcome.

Tasks:

- Identify a list of features that will be constructed
 - Identify which table(s) will be used to construct each feature
- Construct SQL code to build the model training dataset
 - Parameterize your code so you can easily change time periods on the fly, and to make it possible to use one query for both training and scoring the model.
 - Ensure that there is no leakage from your Prediction Window to your Lookback Period. For example, if your Prediction Window is data within the last 90 days, the feature data should only come from data at least 91 days in the past
 - Ensure that your dataset is aggregated/rolled-up to the appropriate level. For example, if you are making a prediction about accounts, the output of your SQL code should be aggregated at the account level.

Considerations:

- Behaviors make for great features. The best predictors about future behavior are often past behaviors.
- Using historical data is a great way to create "change features". For example, instead of just looking at license utilization from one point in time, you can construct multiple features to see how it has changed over time and capture the magnitude, direction, and acceleration of the feature.
- Eventually, the GitLab Data Science team will implement a feature store that will allow to easily creation and retrieval of features to be used for modeling.
- Using a standard naming convention for fields ensures that someone else is able to pickup the and understand the code easily. Consider using the suffixes cnt, pct, amt, flag for fields containing counts, percents, currency amounts, and boolean flags, respectively.

Completion Criteria:

- Generate SQL code that can be run in Snowflake to produce your modeling dataset.
- SQL Code can also be run in Jupyter/Python (this is a requirement for our production pipelines).

3c: Modeling & Implementation Plan

Iterative with 3a & 3b

Sizing: Medium

Purpose: Putting together a modeling plan will allow you to communicate to the stakeholders how you intend to answer their problem statement using data science. A plan should clearly layout any necessary definitions, data sources, methodologies, and outputs. Additionally, constructing a plan will allow for faster, smoother development in future iterations.

Tasks:

In the project issue, document the following:

- A clear, concise, and easy to understand project name. E.g. Propensity to Churn
- Clearly defined and specific outcome/target that is being measured, both in plain English and in syntax. E.g. "Likelihood for a paid account to reduce Annual Reoccurring Revenue (ARR) by at least 10% in the next 12 months". The definition should include the following elements:
 - Population: e.g. Paid Accounts
 - Measurable Outcome: e.g. 10% or greater reduction in ARR
 - Timeframe: e.g. Next 12 Months
- Data sources to be used and specific date windows
- Brief overview of the modeling process, including algorithmic approaches that will be attempted: e.g. logistic regression, XGBoost, MARS
- Model Outputs: e.g. A propensity score (0-100%), top 3 local insights for each account, Inspector and Monitoring dashboards.
- Where the model outputs will reside (Snowflake, Salesforce, etc.) and the DRI for ensuring the source gets populated.
- Project timeline

Considerations:

- Does the definition of the outcome/target align with the intended purpose, outcome, and outputs of the project?
- Does the modeling approach allow you to produce the desired model outputs?
- Do the model outputs allow you to address the overall objective of the project?

Completion Criteria:

- Review plan with stakeholders and modify as necessarily based on feedback.

4: Train MVP Model

Sizing: Large

Purpose: Using the dataset created in the previous stage, prepare, model, and glean insights that directly address the problem statement.

Tasks:

- Conduct a data gut-check to make sure the data looks proper:
 - Naming conventions are consistent
 - Field types look correct (e.g. numeric fields are formatted as numeric)
 - Distribution/Min/Max/Mean/SD look reasonable
 - NULLs for features we expect to have nulls
 - Zero-fills for features we expect to have zero
 - Coverage: are there features where there is not a lot of variation (e.g. mostly zero). If so, you may want to consider creating flag fields for those features instead.
 - Are there clear distinctions between different subsets of the modeling population (e.g. Free vs Paid accounts). If so, it may make sense to split into more than one model if those features are highly predictive of the outcome.
 - Based on the above analysis, does it make sense to create a different view of the potential features (change to bands/deciles/quartiles/etc)?

- Model prep: may include the following components:
 - Outlier detection
 - Missing fill
 - Dummy coding and indicator creation
 - Removing features due to low variation, multicollinearity, high correlation to the outcome, etc.
 - Check logic for possible data leakage
- The Data Science team has created the gitlabds python package to assist in many of these data prep tasks. If you're using the data science jupyter container, this package is already installed.
- Model iteration
 - Logging model training runs/experiments. The Data Science team is currently implementing MLFlow to track model training iterations. More details will be shared when it is available
- Final model candidate selection

Considerations:

- If you would like to use the Data Science Team model production process (see Phase 6), your code will need to be able to execute on the data science jupyter container. Therefore, it is recommended that you use this container to train your model.
 - If your model requires packages that are not installed on the data-science repository, please raise an issue with the Data Science team, and we will add them.
- The Data Science team has a number of tools and resources to assist you in this process. Visit the Data Science Handbook page for additional information. These include:
 - Pre-Configured JupyterLab Container
 - gitlabds python toolset
 - Modeling Process Templates
- Sometimes it's a good idea to build a baseline model that represents the status quo so that there is a tangible solution to compare the model against.

Completion Criteria:

- Test/Validation sample shows high predictability, good discrimination, and low variation compared to the train sample,
- Model lift is monotonic and deciles are roughly equal
- MVP model tells an interesting story based on the predictors (features) present in the model
- One or two predictors are not accounting for the majority of the predictive power of the model

5: Present Model Results and Insights to Stakeholders

Sizing: Medium

Purpose: Synthesize the findings from your model and report back to the stakeholders.

Tasks:

- A presentation should include the following elements:
 - Problem statement and how the model is solving for it.
 - Timeline (where we are in the project)

- Data sources used/considered
- (Optional) Very high-level methodology
- Key features
- How we know it's a good model (e.g. performance metrics, lift)
- Model limitations
- Conclusions
- Next steps

Considerations:

- Although data scientists spend a lot of time (and are excited about!) compiling datasets and iterating over model training runs, that information is best left out of a presentation to stakeholders. Try to keep the deck as non-technical as possible. Any technical information, if desired, can be put into an appendix.
- Same goes for methodologies. It is recommended to limit methodologies to one slide -- mostly so we have documentation of what was done for the project and not because most non-data scientists care :upsidedown:

Completion Criteria:

- Sign-off from stakeholder that the model is an acceptable solution to the problem. This may take several iterations.

6: Operationalize Model

Sizing: Medium

Purpose: In order to score a model independently of a training run -- and on data outside of the training time period -- we need to operationalize a scoring process and add it to the data science production pipeline

Tasks:

- Place all your model parameters into a scoring parameters yaml file.
- Create a new scoring notebook.
- Create a scoring jupyter notebook (a python script also works)
 - Create any hard-codes that are necessary. For example, if you performed outliers for training, use those values in your scoring code (as opposed to calculating new outliers on your scoring dataset)
 - Create dummy codes for any dummy-coded features in your model
 - Make sure you are completing all the same model prep steps in the scoring code as you did in the training code.
- Once you have your scoring notebook ready, a good way to check for errors is to run your training dataset through it. You should arrive at the same model score descriptives and decile breakdowns as you did in your training code.
 - If the numbers are not aligning up, it is most likely due to an issue with your scoring code. The best way to check this is to look at the descriptives of the features. Look at the descriptive right before they get scored in the training code and in the scoring code. If one or more features are off, then you have most likely located the issue.
- Now you can run your code with your scoring dataset. Be sure to update your sql code so it is

parameterized to use the most current data available.

- Examine the model decile distribution of your scored records. They should be roughly equal. If they are wildly off, or if they vary widely in each scoring run, it could be a sign that your model is overfit/it/data-team/fit and cannot generalize beyond the training dataset.
- Add .sql file, parameters.yml, model artifacts, and jupyter notebook to a production directory in your repo.
- Create a new issue using the Scheduling Notebook Request template, follow any additional steps, and tag @gitlab-data/engineers when ready to operationalize

Considerations:

- This step is prone to a lot of human error, so completing the tasks outlined above is helpful to ensure accurate scoring of the model.
- What is the scoring cadence (Daily/Weekly/Month)? Batch or real-time?
 - real-time scoring will likely have different technological requirements than batch scoring.

Completion Criteria:

- When scoring code is run on the training dataset, the score descriptives are identical to what was observed during training
- Scoring data is set to use the most currently available data
- Model deciles are in rough alignment to what was observed in training
- DAG is created for the model pipeline and score populate appropriately

7: Model Monitoring

Sizing: Medium

Purpose: The purpose of setting up dashboards for your model are two-fold: 1) Monitor model performance and lift "in the wild" and; 2) Provide an easy point of access to end-users to consume and understand model outputs. As we migrate to a new data visualization and data observability tools, we hope to streamline, automate, and simplify creating model dashboards.

Tasks:

- Create an "Inspector" dashboard that is targeted towards end-users/consumers of the model outputs.
- Create a "Results" dashboard to track model performance over time.

Considerations:

- This process will hopefully become more streamlined as we shift to a new analytics visualization solution

Completion Criteria:

- Creation and sign-off of dashboards

8: Project Retrospective

Sizing: X-Small

Purpose: The purpose of the retrospective is to help the data science team learn and improve as much as possible from every project.

Tasks:

- Create retrospective issue.
- Invite and solicit feedback from everyone involved in the project (stakeholders, contributors, etc.). Specifically, we want to know:
 1. What praise do you have for the group?
 1. What went well with this project?
 1. What didn't go well with this project?
 1. What can we improve going forward?

Considerations:

- Make sure issue is private (confidential) to the data group, plus anyone else who worked with the group during the project, to ensure everyone feels comfortable sharing freely.

Completion Criteria:

- Gather feedback , synthesize learnings, and close the issue.

SECTION: enterprise-data/organization/internship-experience/_index.md

Purpose

This page is created to share the experiences of Data Team members who completed internships with various teams within GitLab, with the aim of providing insights, learnings, and best practices.

Internship Experience

Lessons from the Trenches: Insights from an SRE Internship

SECTION: enterprise-data/organization/internship-experience/lessons-from-the-trenches-ved-prakash.md

Lessons from the Trenches: Insights from an SRE Internship

Introduction

Embarking on an internship with the Site Reliability Engineering (SRE) foundation team has been a remarkable chapter in my career journey. Over the past few months, I've gained invaluable insights, honed my technical skills, and learned what it truly means to maintain and enhance

the reliability of large-scale systems. My main goal for doing an internship with the SRE foundation team was to understand how they use GitLab itself to manage the whole infrastructure, how to manage the vault and how to adopt the best practices of infrastructure as code to spin a scalable infrastructure. Here, I'd like to share my experiences, challenges, and the lessons I've learned.

What is SRE?

If you need to become more familiar with SRE, it's a discipline that bridges the gap between software engineering, and IT operations. SREs are responsible for building and maintaining highly reliable, scalable, and efficient systems. They use a combination of software engineering skills, automation, and monitoring to ensure that services are always available and performing optimally.

Different teams of SRE within GitLab

Delivery: The Delivery Group enables GitLab Engineering to deliver features safely, scalable, and efficiently to both GitLab.com and self-managed customers.

OPS: The Ops team is an infrastructure team under SaaS Platforms that focuses on improving processes vital to the successful operations of GitLab.

Foundation: Foundations is responsible for several services related to GitLab SaaS Platforms. The Services the Foundations team is responsible for fall into two general categories: Core and Edge. They are primarily responsible for K8s, Config, Service discovery, secret management, CDN, DNS, Load Balancing, and Network.

Scalability: Scalability focuses on observability, forecasting, and projection systems that enable development engineering to predict system growth in their areas of responsibility.

Dedicated: This group of the SRE team is dedicated to a single-tenant GitLab environment, served through a GitLab-dedicated platform.

DBRE: The DBRE team is responsible for anything related to Postgres database installation, upgrade, maintenance, etc. (now part of the platform).

Gitaly Team: The Gitaly team is responsible for building and maintaining systems to ensure that the Git data storage tier of GitLab instances, and GitLab.com in particular, is reliable, secure, and fast.

The process of enrolling into SRE internship

Reach out to the team you want to intern with.

Followed by issue as per the handbook page and get alignment on the time you would like to spend between both your team management and another team manager.

Onboarding in SRE team

The onboarding process for the SRE team was comprehensive and well-structured, designed to integrate new interns into the team's workflow quickly:

1. **Introduction to Team and Tools:** I was introduced to team members and given an overview of the tools and technologies used in our SRE practice. This included familiarization with our monitoring systems architecture overview.

2. **Access and Permissions:** The team ensured I had access rights to relevant systems and

repositories, allowing me to start contributing from day one.